Cardinality-Constrained Portfolio Optimization

A Genetic Algorithm Out-of-Sample Evaluation

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June 2026

Table of contents

01

Motivation & Research Questions

Motivation

THE PROBLEM

- Real portfolios cannot hold 500+ stocks efficiently.
- Transaction costs, complexity, and regulation matter.
- Markowitz MVO ignores these practical constraints.

THE GAP

- Cardinality constraints make the problem NP-hard.
- Standard QP solvers cannot handle it, metaheuristics are needed.
- GAs are a natural fit, but out-of-sample evidence remains limited.

Can a GA beat simpler benchmarks across 20 years of real market data, and if not, why not?

Research Questions

RQ1 What is the impact of a 0.15 maximum weight bound on out-of-sample risk-adjusted performance vs unconstrained MVO?

RQ2 Do evolutionary algorithms achieve competitive Sharpe ratio vs constrained MVO, unconstrained MVO, and 1/N under realistic constraints?

RQ3 How does the choice of cardinality K affect GA Sharpe ratio, turnover, and concentration, and how does adaptive K compare to fixed-K configurations?

02

Algorithm & Experimental Setup

Algorithm

Why a Genetic Algorithm?

NP-hard problem - QP solvers fail. Real-valued chromosome fits portfolio weights better than ES or DE. Chang (2000), Streichert (2003).

Chromosome Representation

N ≈ 867 positions, K ∈ [10, 30] non-zero entries each ∈ [0.02, 0.15], Σw = 1

0	0.08	0	0.15	0.05	0	0.10	...	0	0.09	0.10	0.02	0	0.04	0

Live Walkthrough

Steps: population -> selection -> crossover -> mutation -> repair + local refinement -> new generation (elitism + median-fitness run selection)

Experimental Setup

Data

- CRSP US equities, Jan 2000 - Dec 2025
- Approximately 867 eligible stocks/month (NYSE + NASDAQ, market cap ≥ $2B)
- 252 out-of-sample rebalancing periods (Jan 2005 - Dec 2025)
- Risk-free rate: FRED 3-month T-bill (DTB3)

CRSP

FRED
ECONOMIC DATA | ST. LOUIS FED

Setup

- **Estimation window:** 60 months rolling
- **Covariance:** Ledoit-Wolf shrinkage
- **Optuna tuning:** 2005–2012, 15 Trials, TPE

Experimental Setup

Benchmarks

Constrained MVO: weights ≤ 15%

Unconstrained MVO: weights ≤ 100%

1/N: equal weights for ~867 stocks

Costs

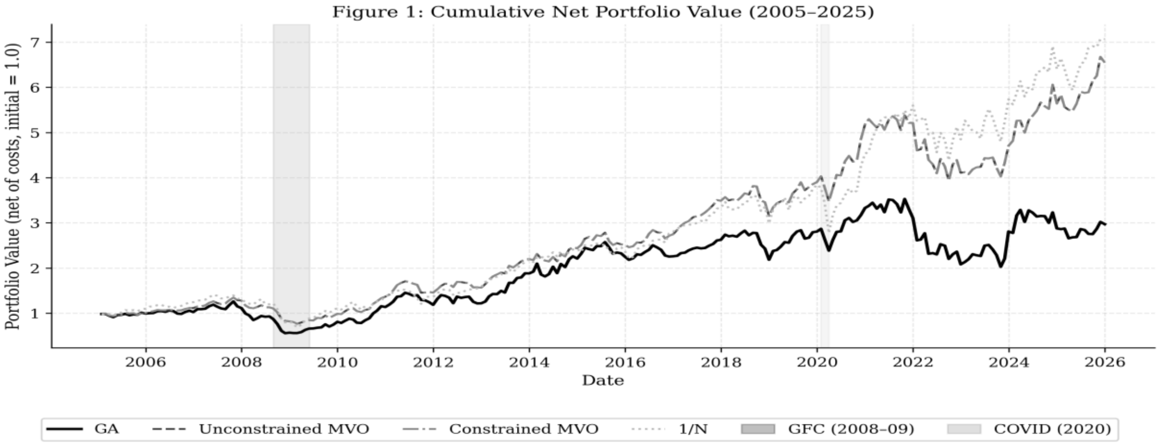
Transaction cost γ = 0.3% on all models

Model	Weight bounds	Cardinality	Estimation
GA	[0.02, 0.15]	$K \in [10, 30]$	Yes
Constrained MVO	[0.00, 0.15]	None	Yes
Unconstrained MVO	[0.00, 1.00]	None	Yes
1/N	1/N each	All ~867	No

03

Results

Cumulative returns



All returns net of transaction costs (γ = 0.3%). Initial portfolio value normalised to 1.0. Shaded regions indicate the Global Financial Crisis (Sep 2008 – Jun 2009) and COVID-19 shock (Feb – Apr 2020).

By December 2025 both MVO variants and 1/N reach **higher** terminal value than the GA.

Main performance table

Metric	GA	Const. MVO	Unconst. MVO	1/N
Sharpe (net)	**0.2741**	0.5810	0.5809	0.5307
Ann. Return	5.46%	8.28%	8.28%	9.13%
Ann. Vol.	19.92%	14.26%	14.26%	17.20%
Max DD	−55.76%	−43.07%	−43.07%	−51.31%
Turnover	22.19%	17.07%	17.07%	4.50%

The GA **underperforms** all three benchmarks. Both MVO variants reach nearly identical Sharpe ratios, the 0.15 weight cap barely binds across 867 stocks.

Statistical significance

Paired t-test (mean return difference)
Jobson-Korkie test with Memmel correction (Sharpe ratio difference)

Comparison	Sharpe Diff	JK p-value	JK Sig.	t p-value	t Sig.
GA vs 1/N	−0.257	0.1316	no	0.2558	no
GA vs Unconst. MVO	−0.307	**0.0045**	**yes**	0.2051	no
GA vs Const. MVO	−0.307	**0.0045**	**yes**	0.2051	no

The Sharpe underperformance against both MVO variants is **statistically significant** (JK, $p < 0.01$), but the mean return difference is not significant for any benchmark.

The central finding: λ-ablation

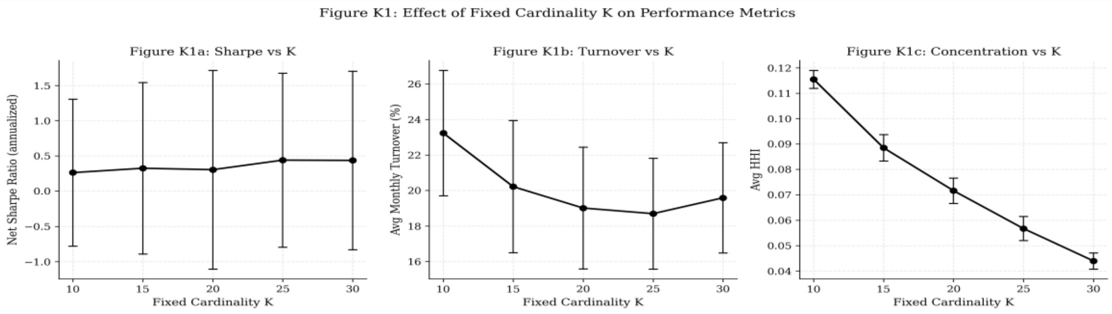
Full 252-period re-run with penalty removed ($\lambda = 0$) vs tuned ($\lambda = 1.8437$).

	Tuned λ = 1.8437	λ = 0 (no penalty)
Net Sharpe	0.2741	**0.4993**
Avg cardinality $\bar{K}$	16.3	25.1
Avg turnover	22.2%	57.1%

Removing the penalty nearly doubles Sharpe, the penalty suppresses cardinality too **aggressively**, costing more in return than it saves in trading costs.

Fixed-K sensitivity

Fixed hyperparameters held constant, only K varies across {10, 15, 20, 25, 30}.



Effect of fixed cardinality K on net Sharpe ratio, average monthly turnover, and portfolio concentration (HHI). All models use fixed hyperparameters (pc=0.6054, pm=0.1370, σ_m=0.1469, λ=1.8437) to isolate the effect of K. Error bars show ±1 std of the rolling 12-month estimates across periods.

Sharpe rises with K up to K = 25–30, while concentration falls monotonically. The adaptive GA settles at K = 16.3, below every fixed-K configuration except K = 10.

04

Conclusions & Future Work

Conclusions

RQ1 (constraints vs unconstrained MVO): Weight constraints have negligible impact, constrained and unconstrained MVO reach near-identical Sharpe (0.5810 vs 0.5809) across 867 stocks.

RQ2 (GA vs benchmarks): GA underperforms all benchmarks (Sharpe 0.2741 vs 0.53-0.58, JK p=0.004 vs MVO). Root cause: over-aggressive turnover penalty, removing it raises Sharpe to 0.4993.

RQ3 (cardinality K): Higher K improves Sharpe (peaks K=25–30), concentration falls monotonically. Turnover penalty suppresses adaptive K to 16.3, below all fixed-K configs except K=10.

Key takeaway: The main bottleneck is the turnover penalty, not the evolutionary approach.

Future Work

Walk-forward Optuna tuning

Extend to international or multi-asset portfolios

Tune λ per fixed-K configuration

Scale to 30 parallel GA runners

05

Threats to Validity

Threats to Validity

Internal

8 runners instead of 30, relaxed MVO solver tolerances, 105 tests validate implementation.

External

US large/mid-cap only, GFC and COVID period not representative of average conditions.

Construct

Sharpe assumes normality, tuning window overlaps evaluation (mild data-snooping).

Conclusion

JK rejects GA vs MVO (p=0.0045), paired t-test does not (p=0.205), tests answer different questions.

Thank you!

Do you have any questions?

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github.com/georgeded/ga-portfolio-optimization